Characterization and Differentiation of High Energy Cyclic Organic Peroxides by GC/FT-IR, GC-MS, FT-IR and Raman Microscopy

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ABSTRACT

The characterization of Tetracetone Tetraperoxide (TRATRP), Triacetone Triperoxide (TATP), Diacetone Diperoxide (DADP), Tricyclohexylidene Triperoxide and Dibenzo Diperoxide using GC-MS, GC-FTIR, FTIR, FT-NMR and Raman Spectroscopy is reported. These compounds were synthesized, purified and characterized in the laboratory in order to develop methodologies for their trace detection. During this study, TATP has been synthesized by different methods obtaining high purity and good yields, even using common household products. DADP synthetic routes reported in the literature were verified. The methods described, including those that produce mixtures with TATP and other peroxides forms were also tested. This study will also focused in the preparation of other cyclic peroxides, including Hexamethelene Triperoxide Diamine (HMTD) and different forms of cyclic peroxides from ketones. This issue of thermodynamic versus kinetic control of secondary products of all syntheses and the effect of temperature in the distribution sub products of the syntheses was also addressed. A vibrational differentiation study of was carried out. Differences were found computationally in the ν(O-O), ν(C-O), δ(CH_3-C) and δ(C-O) for Raman and IR bands and retention time and fragment patron for GC-MS and GC-FT-IR.

Keywords: TATP, Raman and FTIR, GC-MS and DFT

1. INTRODUCTION

In recent decades, cyclic organic peroxides have been regaining their lost interest due their increased use as explosive materials by terrorists. Since their initial syntheses by Wollfenstein[1] in 1895, their instability was accounted for and few groups tried to work with them decades later.[2] In the last few years, cyclic peroxides began to appear in several terrorist cases.[3-6] Several federal agencies discovered that substances such as Triacetone Triperoxide (TATP), not only have a simple synthesis,[7] but current explosive detection technology did not detect them,[3, 8] Many researchers have developed studies to find specific methods to detect them. Buttigieg presented IMS as a solution for the detection of TATP and Hexamethylene Triperoxide Diamine (HMTD).[9] Others like Widmer and coworkers, have develop methods for HPLC detection.[10] Recently, Zeiri proposed a method for the detection of TATP using metals.[11] In the year 2000, work was begun in the lab with TATP and DADP to develop methodology for trace detection.[3,4] These early studies confirmed the use of GC/MS for trace detection and separation of peroxide from mixtures.[3]

This study presents the characterization of DADP, separated and in mixtures with TATP and documentation of synthesis and characterization of Peroryacetone Tetramer.[12] Other peroxides were also synthesized which produced common Raman peaks with TATP or DADP. The first GC/FT-IR report of organic peroxides will be shown.

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2. METHODS

2.1. Reagents

The chemical reagents used in this research were: acetone, benzaldehyde, acetophenone, cyclobutanone, cyclopentanone, cyclohexanone, (98% w/w, Aldrich-Sigma Chemical Co., Milwaukee, WI), hydrogen peroxide (H_2O_2, 50% in water, Aldrich-Sigma), and hydrochloric acid (HCl,12 M, Merck, VWR International, Inc., West Chester, PA 19380), and methanosulfuric acid (HCH_3SO_3, Aldrich-Sigma).

2.2. Instrumentation

Fourier Transform Infrared (FTIR) spectrometer used was a Bruker Optics model IFS 66v/S. Raman instrumentation used included Bruker Optics Fourier Transform Raman microscope and macro model RFS 100/s, equipped with 1064 nm Nd:YAG laser, Renishaw RM2000 Raman Microscope equipped with Renishaw high power solid state diode laser operating at 785 nm and Jobin-Ybon T64000 triple Raman monochromator with 514.5 nm Coherent, Inc. INNOVA 308 Ar^+ laser system. Dispersive Raman equipment operated in confocal mode. All spectrometers were equipped with polarization analyzers and de-scramblers. For Nuclear Magnetic Resonance characterization, a Varian Gemini 300 MHz FT-NMR was used for ^1H and ^{13}C analysis. A Hewlett-Packard gas chromatograph model GC-MS HP 5890 GC system coupled to a HP 5790 mass selective detector was used for the mass spectroscopy experiments. A Perkin-Elmer System 2000 GC/FT-IR was used to analyze the peroxides mixtures in gas phase.

2.3. Method of Analysis

The characterization of DADP and other peroxides was made by various techniques such as Gas Chromatography-Mass Spectrometry (GC/MS), Fourier Transform-Infrared Spectroscopy (FTIR), Gas Chromatography-Fourier Transform (GC/FT-IR), Fourier Transform-Nuclear Magnetic Resonance (FT-NMR) and Raman Spectroscopy, to confirm purity and obtain positive identification of the compounds. The NMR studies (^1H-NMR, ^{13}C-NMR and attached proton test, APT) were performed in a Varian Gemini, 300 MHz FT-NMR with 10-20 mg of sample diluted in $CDCl_3$. Atlantic Microlabs performed the elemental analysis of TATP to confirm the empirical formula. The GC-infrared spectroscopic studies were performed with a Perkin Elmer System 2000 GC/FTIR equipped with MID and Far IR detectors which can be used independently or coupled to the GC. In the FTIR analysis, the sample was prepared in a 5% in a mixture with KBr. For GC/FTIR analysis, concentrated solutions in methylene chloride and chloroform as solvents were injected (3 μL) in split mode. The mass spectra of the organic peroxides were performed in the HP GC-MS injecting 0.1 μL of a slightly diluted solution of the peroxide compounds. The GC parameters were adapted to prevent thermal decomposition.

2.4 GC Method

GC/FT-IR program method used was: injector temperature 120°C, initial column temperature, 80°C for 10 min, ramp 30°C/min to 100°C for 25 min, 130°C at 40°C/min for 10min and reaching a final temperature of 180°C, detector temperature 200°C.

3. RESULTS AND DISCUSSION

3.1. Synthesis

The synthesis of the peroxides, following preparation of TATP, was made with the specific ketone and hydrogen peroxide in a 1:1 molar ratio (2 mL ketone mixed with 2 mL H_2O_2) in a cold system catalyzed by a strong acid (HCl, H_2SO_4 or HCH_3SO_3). The ketones and aldehyde were cooled to between 0°C and -5°C with acid, in pure form and in dichloromethane solution. Addition of hydrogen peroxide was done drop wise to prevent violent reactions.* White crystalline products were obtained, filtered and washed with distiller water. Recrystallization of the products was then carried out in dichloromethane or ether. Syntheses at 20 ^{0}C produced a mixture of dimer, trimer and even

* This reaction can produce large amounts of heat that may result in the detonation of the organic peroxides formed, leading to micro explosions that contaminate the products.

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tetramer of cyclic peroxides. Figure 1a shows the generally accepted mechanism of synthesis. The preparation of DADP was more complex than TATP. The synthesis of the organic peroxide was tried by several methods. Most of these produced mixtures of peroxides including TATP and DADP. Few syntheses successfully produced principally DADP. The product in crude form of the successful experiments sublimates faster than they could be characterized by most of the experiments performed. Figure 1b shows a white light micrograph of TATP crystals obtained.

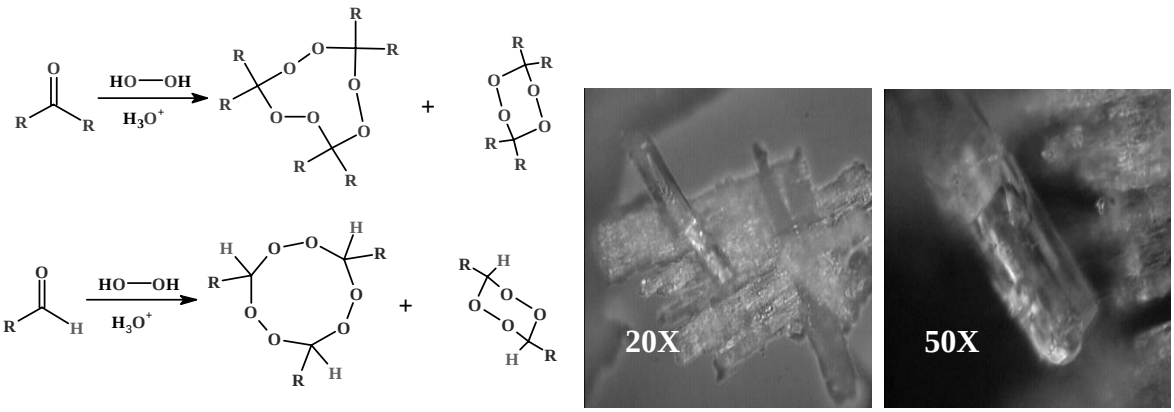
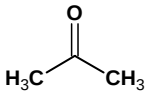
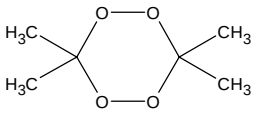
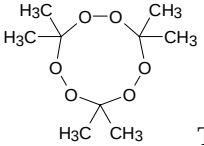
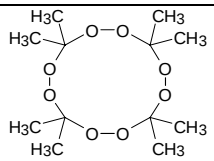
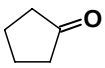
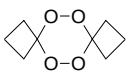
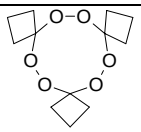
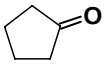
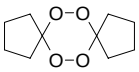
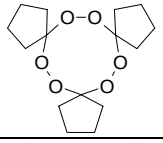
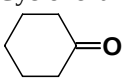
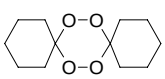
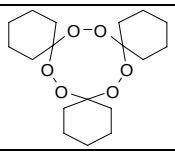


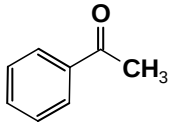
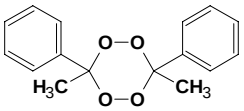
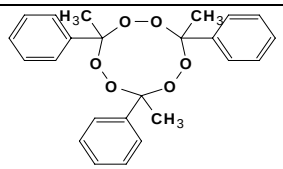
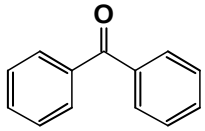
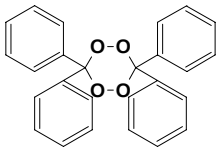
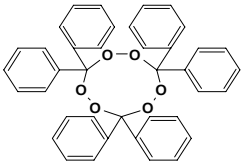
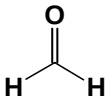
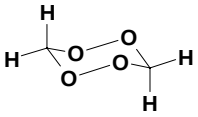
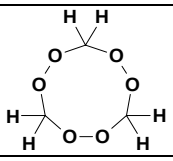
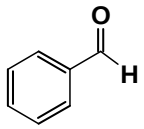
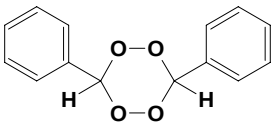
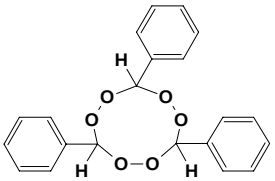
Figure 1. (a) Synthesis of ketone and aldehyde peroxides; (b) white light micrographs of TATP crystals

The other cyclic organic peroxides were made from benzaldehyde, cyclobutanone, cyclopentanone, cyclohexanone, acetophenone, and benzophenone. Table 1 shows the peroxides that have been synthesized at present and the status of preparation for the others proposed.

Table 1. Synthesis of ketone and aldehyde peroxides

Primary Reactant	Possible and Expected Products		
Ketones	**Dimer**	**Trimer**	**Tetramer**
Acetone 	 DADP	 TATP	 TRATRP
Deuterated Acetone	In Process	TDATP	
Cyclobutanone* 			Observations:
Cyclopentanone 			Observations: White crude solid
Cyclohexanone 			Observations: A very viscous (pasty) transparent liquid.

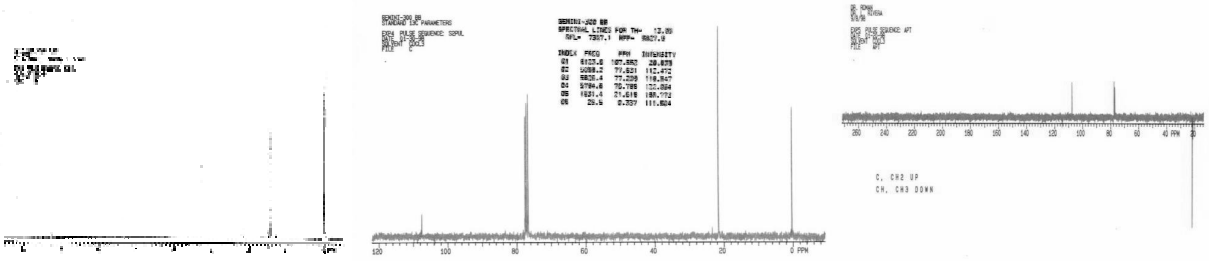
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Acetophenone 			Observations: White small crystals
Benzophenone 			Observations: White sharp crystals. Benzophenone required to be diluted or pulverized
Aldehydes			
Formaldehyde* 			Observations: For safety reasons this peroxide was synthesized in solution.
Benzaldehyde 			Observations: White crude solid that transforms in laminar transparent crystals.

3.2 Nuclear Magnetic Resonance Analysis (FT-NMR) of Peroxyacetones

Figure 2 shows the characterization of TATP and DADP by ^{1}H-NMR and ^{13}C-NMR. As shown on the figure, there are substantial differences between both compounds.

^{1}H-NMR, ^{13}C-NMR and ^{13}C-NMR APT of TATP



^{1}H-NMR and ^{13}C-NMR of DADP

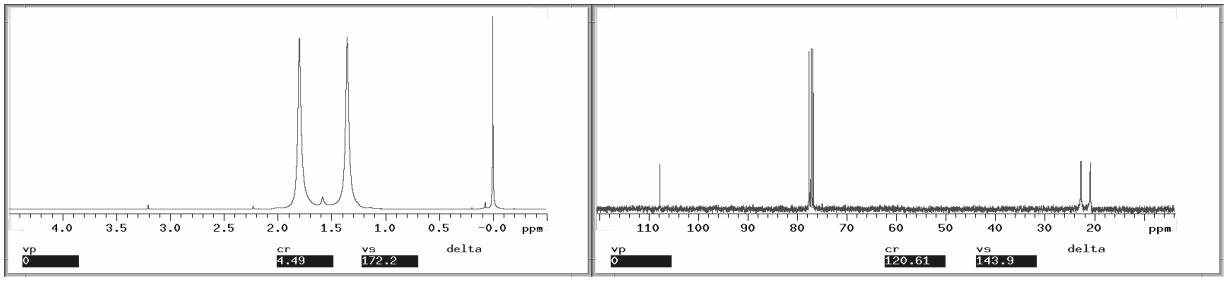


Figure 2. FT-NMR spectra of TATP, DADP and Attached Proton Test (ATP) of TATP.

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TATP proton analysis reveals a strong singlet at 1.45 ppm that belongs to methyl groups of the acetone-like part of the molecule. Due the absence of hydrogen atoms close, the inductive effect caused by both oxygen bonds attached to the alpha carbon only produced a singlet that is shifted down field to 1.45 δ when compared to those of methyl groups of alcohols (1.17 δ). The proton FT-NMR spectrum of DADP shown in Figure 2 revealed two strong symmetric singlets at 1.36 δ and 1.80 δ that belong to methyl groups of the DADP molecule. The two sharp absorptions separated by a chemical shift of 0.44 δ demonstrate the presence of axial and equatorial methyl groups in DADP caused by a cyclohexane-like conformation. This conformation has also been proposed for DADP in literature.[5] The inductive effect, produced by both oxygen bonds attach to the alpha carbon, generates only singlets of the beta carbons that were moved down field to 1.36 δ and 1.80 δ in comparison to methyl groups of alcohols is which appear at 1.17 δ. These chemical shifts of DADP were close to those observed for TATP at 1.45 δ that is shown in Figure 2.

The ^{13}C-NMR spectrum shown Figure 2 confirms the structure by revealing peaks of two different types of carbon atoms in the molecule. A singlet at 21.6 δ reveals the presence of the methyl groups of TATP since this large signal is due to the fact that there are six equivalent methyl groups in the TATP molecule. A triplet at 77.2 is a deuterated chloroform signal and the 107.55 δ are signals arising from the three carbon atoms of TATP attached to the oxygen atoms, which explains their down field position. The calculated ^{13}C chemical shifts for both carbons in TATP were estimated in 21.0 δ and 105.7 δ, respectively. For the DADP structure, two singlets at 20.8 δ and 22.7 δ can be attributed to the methyl groups of DADP in the axial and equatorial positions of the molecule. In this case it differed significantly from TATP spectrum since the 107.7 δ shift produced by the oxygen atoms attached to the carbons, is now at same intensity level of the other two carbons. This phenomenon explains that the DADP molecule contains the same proportions of carbon atoms in the molecule, two of each type in the case of DADP: two axial, two equatorial, and two quaternary. The strong triplet at 77.2 δ is a signal attributed to the deuterated chloroform solvent. These conclusions were confirmed when an attached proton test (APT) shown in the Figure 2. The APT demonstrates that the 21.6 δ carbon atoms invert while 107.55 δ were normal. This experiment confirms that the small downfield peak of 107.55 δ is from quaternary carbons and the large upfield peak of 21.6 δ is due methyl carbon atoms of the molecule. The NMR data presented above is a strong evidence of presence of TATP in the sample and supports the FT-IR and Raman spectroscopy data.

3.3 Mass Spectral Analysis of Peroxyacetones

Figure 3 shows the Mass Spectra characterization of TATP and DADP. As shown, the molecular peaks of both compounds are present: 222 m/z for TATP and 149 m/z for DADP. Another peroxide mixture analyzed showed a signal at 296 m/z which could be attributed to the tetramer structure. All these peroxyacetones produced a common fragmentation pattern: base peak of 43 m/z, peaks at 58, 59, 75, 101, 117 m/z. TATP presented a characteristic peak at 89 m/z, in DADP this peak is 91 m/z, and for the tetramer (TRATRP) the mass spectrum also exhibits peaks at 91, 133, and 148 m/z.

3.4 Vibrational Spectroscopic Analysis of Acetone Peroxides

Raman spectroscopy was a useful tool in this characterization, especially when the experiments were run in the fingerprint region. Figure 5 presents the Raman spectra of TATP, DADP and a mixture of peroxyacetones in a Jobin-Ybon T64000. Raman shows a strong doublet band, at 3011 cm^{-1} and 2996 cm^{-1} that belongs to asymmetric stretching of the CH_3 but the doublet can only be explain as a population problem due the six-methyl groups moving at different phases. The characteristic peroxide bands are between 950 cm^{-1} and 750 cm^{-1}, these bands are also present in FT-IR spectra, but these ones have stronger intensities. In DADP, the same bands were presents but the complete spectrum was unable to reproduce because the compound sublimates to fast. Although the mixtures of peroxides were possible to separate by Gas Chromatography, Raman microscopy was used to analyze the liquid solution reacting mixture. As shown in figure 5, all other peroxyacetone spectra contain bands between 950 cm^{-1} and 750 cm^{-1}, even in mixture conditions. The only difference between TATP and DADP is the amount of peaks, DADP produce a simpler spectrum at this region. The mixture spectrum included TATP, DADP and the AP tetramer (TRATRP) which explains the complexity of the ν(O-O) region.

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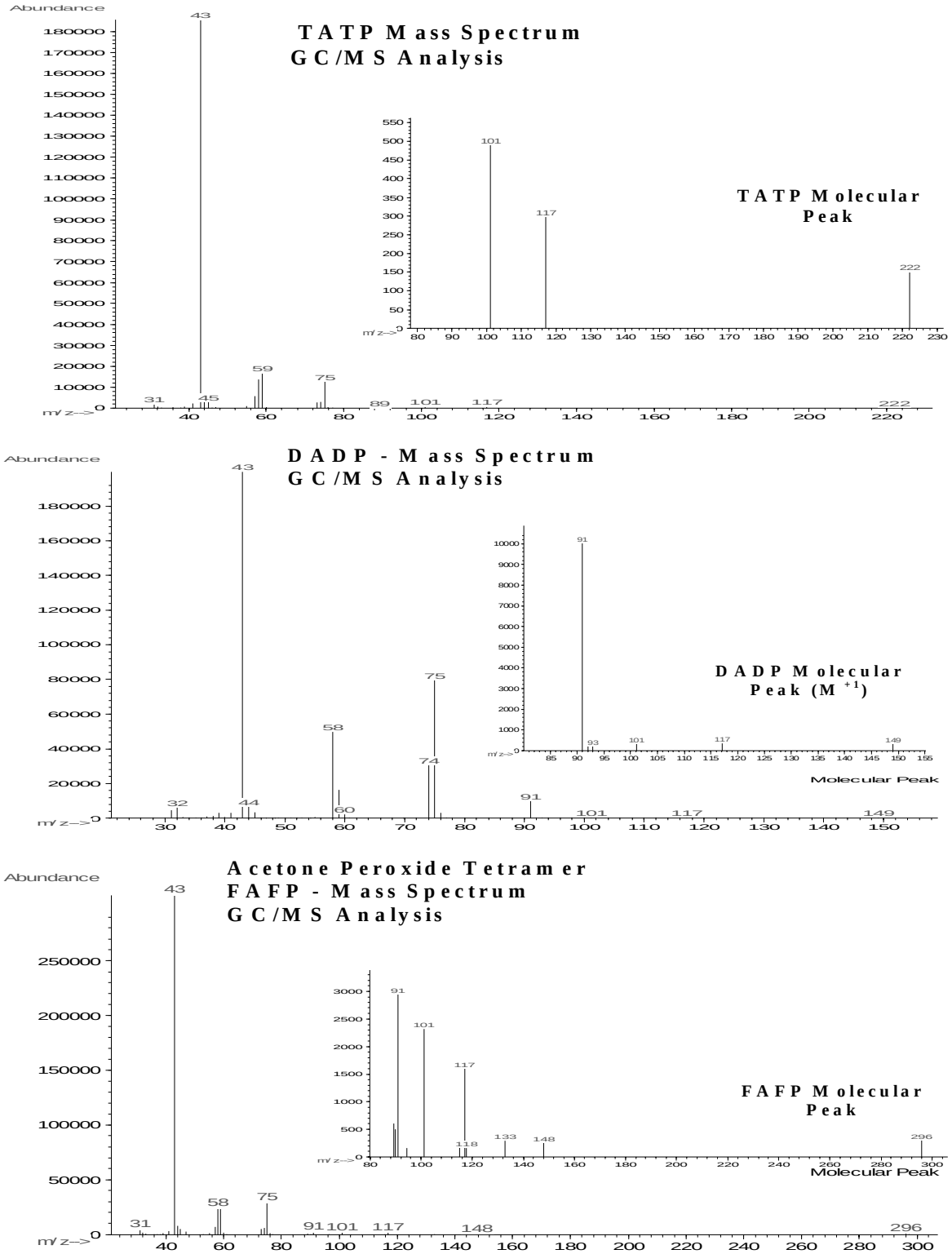


Figure 3. GC-MS spectrum for TATP, DADP and TRATRP

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Raman spectroscopy was a useful tool in this characterization, especially when the experiments were run in the fingerprint region. Figure 5 presents the Raman spectra of TATP, DADP and a mixture of peroxyacetones in a Jobin-Ybon T64000. Raman shows a strong doublet band, at 3011 cm^{-1} and 2996 cm^{-1} that belongs to asymmetric stretching of the CH_3 but the doublet can only be explain as a population problem due the six-methyl groups moving at different phases. The characteristic peroxide bands are between 950 cm^{-1} and 750 cm^{-1}, these bands are also present in FT-IR spectra, but these ones have stronger intensities. In DADP, the same bands were presents but the complete spectrum was unable to reproduce because the compound sublimates to fast. Although the mixtures of peroxides were possible to separate by Gas Chromatography, Raman microscopy was used to analyze the liquid solution reacting mixture. As shown in figure 5, all other peroxyacetone spectra contain bands between 950 cm^{-1} and 750 cm^{-1}, even in mixture conditions. The only difference between TATP and DADP is the amount of peaks, DADP produce a simpler spectrum at this region. The mixture spectrum included TATP, DADP and the AP tetramer which explains the complexity of the ν(O-O) region.

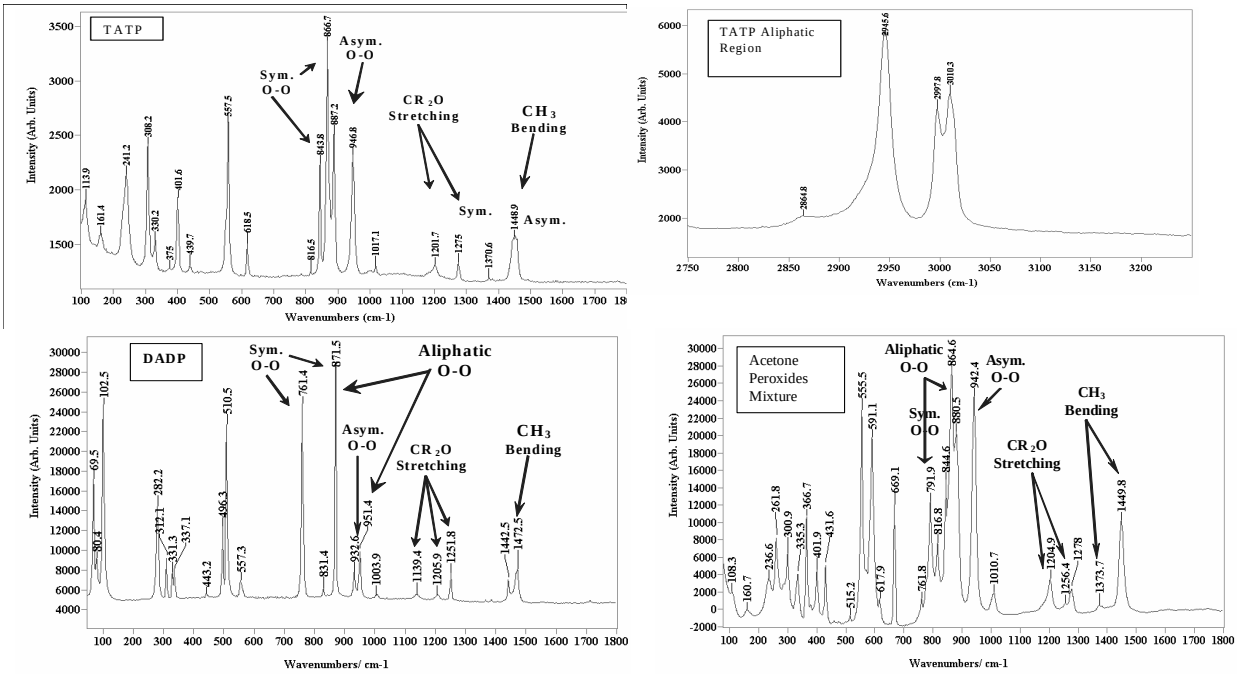


Figure 5. TATP, DADP and peroxide mixture fingerprint spectra in Raman microscope of Jobin-Ybon T64000.

The first peaks shown in figure 5c (DADP) are two medium bands at 1472 and 1443 cm^{-1} that belonged to the asymmetric bending of the methyl groups, followed by the symmetric bending at 1259 (m) and 1209 cm^{-1} (w). A weak band at 1140 cm^{-1} is due the C-C-O and C-O interactions. At 1008 cm^{-1} similar to TATP but at 1016 cm^{-1} the deformation bands, rocking and wagging of the CH_3 could be observed with low intensities because its lack of symmetry. At 951 and 933 cm^{-1}, two medium bands are evident and they were assigned to the asymmetric vibrations of the aliphatic O-O bond. At this frequency, TATP showed only one band. DADP produced a doublet due the structure of the chair conformation or the proximity of both groups in the molecule. A strong peak at 871 cm^{-1} indicates the symmetry of the molecule when the O-O generates a symmetric interaction at this point.

GC/FT-IR analyzed the peroxide mixture, because Raman and FT-IR couldn't determine to which spectra corresponded the bands, and these techniques allow knowing structural information of the other peroxides as well as the mixture was separated. The NMR and the Raman studies were performed to the DADP without purification therefore the studies revealed a substantial purity that permitted us to fully characterize the compound. For GC-MS

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the purity of DADP was not necessary to characterize it. In GC-MS, four principal compounds including TATP and DADP were detected in the more complex DADP synthesis products. In GC/FT-IR, TATP could be identified by its spectrum, but the instability of DADP and the other peroxides complicated their analysis. However, these GC/FT-IR data will be presented at figure 6. The first compound in the chromatogram is the solvent dichloromethane. At 14 minutes, DADP was detected followed by TATP at 27 minutes of elution. After 30 minutes, two acetone peroxides also viewed on the GC/MS results appear, but due the high temperature of the method at this moment, both compounds began to decompose. The shown spectrum is the peroxide labeled as the acetone peroxide tetramer (TRATRP) right after the TATP spectrum.

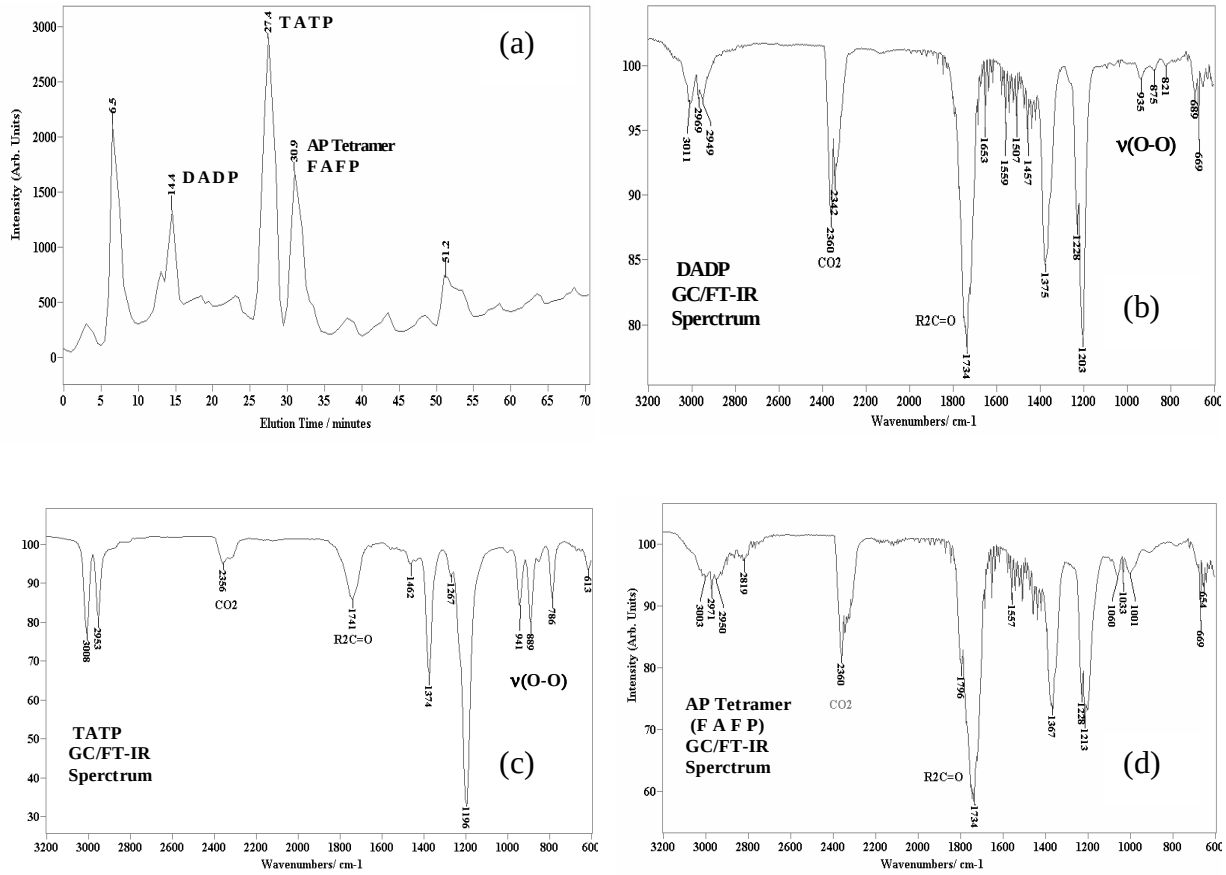


Figure 6. GC/FTIR of Peroxyacetone mixture form DADP synthesis: (a) Chromatogram, (b) DADP, (c) TATP, (d) TRATRP

Figure 6 shows the FTIR spectra that suggest that the compounds synthesized were different from TATP and that most of the peaks agreed with its IR spectra without the problem of decomposition.[13] The multiple and strong bands shown by TATP between 950 and 800 cm^{-1}, where the aliphatic O-O stretching were observed, in the DADP spectrum shown in Figure 6c were reduced to three medium size bands at 935, 875, and 821 cm^{-1}. However, not all of the IR data can be confirmed by the GC/FT-IR analysis because the compound decomposes significantly in the light-pipe transfer-line to the detector. We can conclude that the strong bands at 2360 and 1734 cm^{-1} belong to CO_2 and/or the formation of organic acids, water vapor or ketones. In view of this, it is concluded that the synthesized compounds were cyclic acetone peroxide that was very sensitive to heat. This is a characteristic of DADP, which is supposed to be more sensitive to thermal degradation than TATP. Also, TRATRP revealed that property as well as

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the other acetone peroxide. Most of the DADP IR spectra coincided with Raman spectra, but a third peroxide band present in IR at 831 cm^{-1} is weak in Raman and is part of the last interactions observed in the aliphatic O-O group of the molecule.

In Figure 7, the acetone peroxide tetramer Raman fingerprint spectrum is shown. These crystals were analyzed by GC/MS and presented a high purity. Four-acetone four-peroxide has a very similar spectrum of TATP, perhaps the presence of more important groups create this similarity, but because TRATRP can be also be produce during TATP synthesis, it does not create limitations for peroxyacetone detection by Raman.

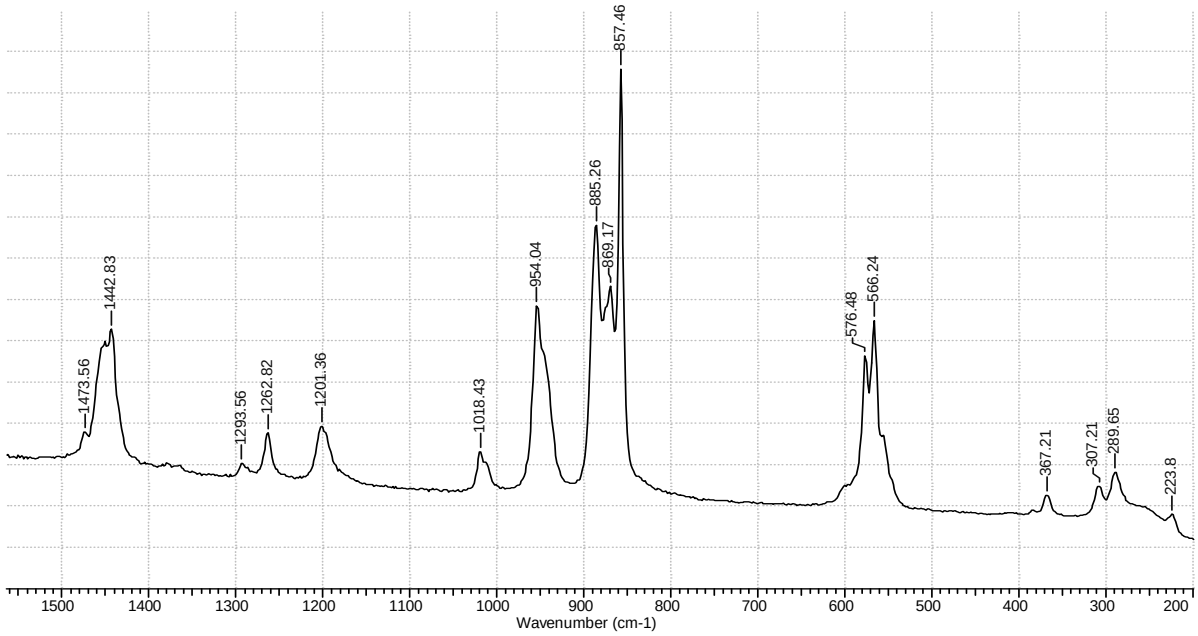


Figure 7. Raman spectra of Four Acetone Tetracetone Tetraperoxide (TRATRP) crystals

3.5 Vibrational Spectroscopic Analysis of other Cyclic Organic Peroxides

Other cyclic peroxides synthesized also showed the ν(O-O) region bands as well as peroxyacetones. Figure 8 shows the obtained Raman spectra of TATP and Cyclopentanone Peroxide. The spectra reveal that the most prominent bands are assigned to ν(O-O). This band shows as low intensity in the Raman spectrum but has high intensity in the IR spectrum, thus indicating its high symmetry nature.

All the peroxides were synthesized following a similar process as the described for TATP synthesis, for the same reason that the compounds expected to be trimeric forms of the peroxides, but previous GC analyses revealed the presence of impurities and mixtures similar to those shown in DADP synthesis. The Raman spectra of trimeric forms of cyclopentanone peroxide and cyclohexanone peroxide revealed some similarities, but the difference in appearance helped to determine which peroxide was analyzed. Figueroa tried to perform a synthesis for formaldehyde, but the peroxide exploded destroying all the equipment.[4] For that reason, our next aldehyde to be synthesized was benzaldehyde peroxide. As shown in Figure 9, Benzaldehyde Peroxide presents one sharp bands at the peroxide region ν(O-O) at 933 cm^{-1} and 808 cm^{-1} in FTIR and 794 cm^{-1} in Raman, this can be explained as symmetric for of the benzaldehyde peroxide or a dimeric form of the synthesized compound. NMR was used for characterization, but impurities of the sample prevented a conclusive analysis.

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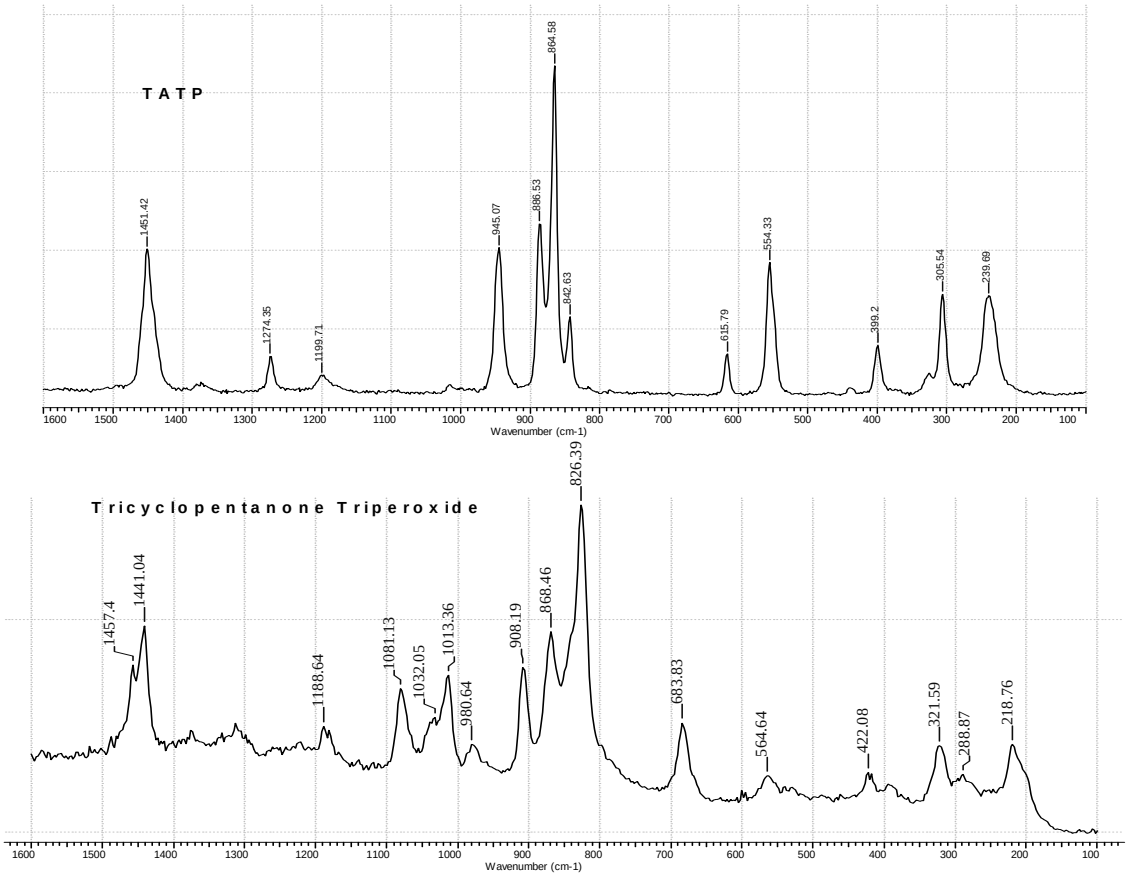


Figure 8. Raman spectra of TATP and Tricyclopentanone Triperoxide in the Renishaw RM2000 Raman Microscope.

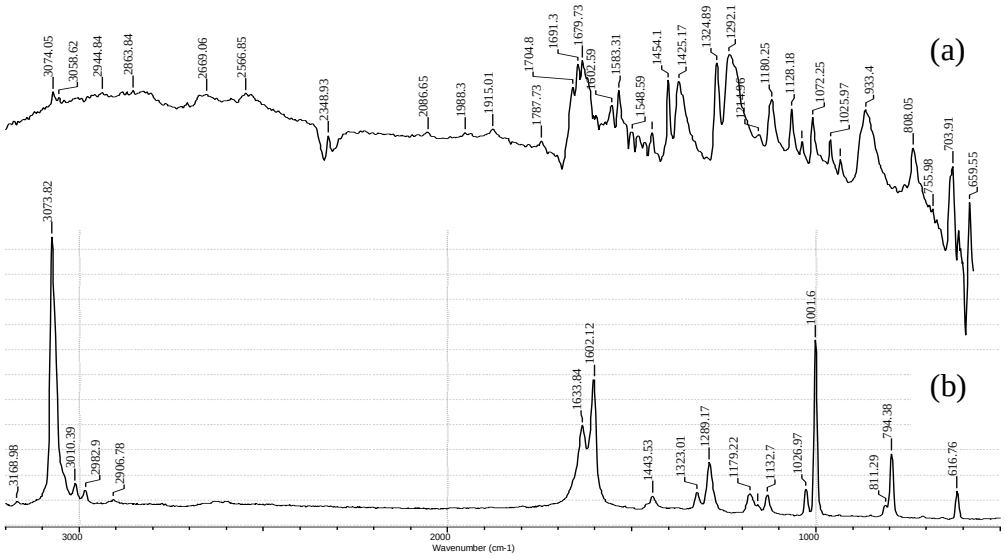


Figure 9. Benzaldehyde Peroxide FTIR (a) and Raman Microscopy (b) spectra.

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4. CONCLUSIONS

As part of this work several cyclic organic peroxide analogous to triacetone triperoxide (TATP) and diacetone diperoxide (DADP) were synthesized in a simple and direct way with the purpose of studying their spectroscopic characterization using vibrational spectroscopy. Previous reports indicate that TATP and DADP mixtures form when the acid catalyst is not strong enough. This observation is true but because the temperature in the solution rose when stronger acids are used. When the temperature was kept below 5°C only one peroxide was formed: TATP. If the temperature rose above 10°C, mixtures of TATP and DADP were formed and even the acetone peroxide tetramer was produced. Also, prolonged contact time of the reacting mixture strong acids generated cyclic peroxides mixtures, such as tetrameric form of acetone peroxide: Tetracetone Tetraperoxide (TRATRP).

Raman and FTIR Microscopy were used as main experimental tools for the spectroscopic characterization of samples. DADP sublimation pressure is so high that it prevented typical spectroscopic measurements. But tetramer TRATRP has a low vapor pressure that can be used to purify this compound and allow the analysis by different techniques. GC-FTIR was proven as a robust technique for cyclic organic peroxides and a complementary technique to more established methods of analysis such as GC-MS. A method of analysis of acetone peroxides such as TATP and DADP was described and can be used for other peroxide mixtures, but the lack of sensitivity limits their capacity for trace detection.

Seven different cyclic organic peroxides were synthesized and characterized by Raman Microscopy, including TATP. A persistent Raman signature of these transparent energetic compounds is the sharp ν(O-O) vibration. This is a distinctive band for all cycloperoxides and can be used to determine if a dimmer or a trimer of the peroxides is formed in a synthesis. The existence of peroxide mixtures was probed experimentally using GC-MS and GC-FTIR. A cyclic organic peroxide was synthesized from an aldehyde (benzaldehyde) and was characterized by NMR, Raman and FTIR as peroxide group member. All the data presented supports that the compound has a complex structure or needed to be further purified to characterize the structure.

5. ACKNOWLEDGEMENTS

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